HANDS-ON MACHINE LEARNING · PROJECT 2026-27

Predicting Online Shoppers' Purchasing Intention

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Data: UCI Online Shoppers Purchasing Intention dataset
archive.ics.uci.edu/dataset/468

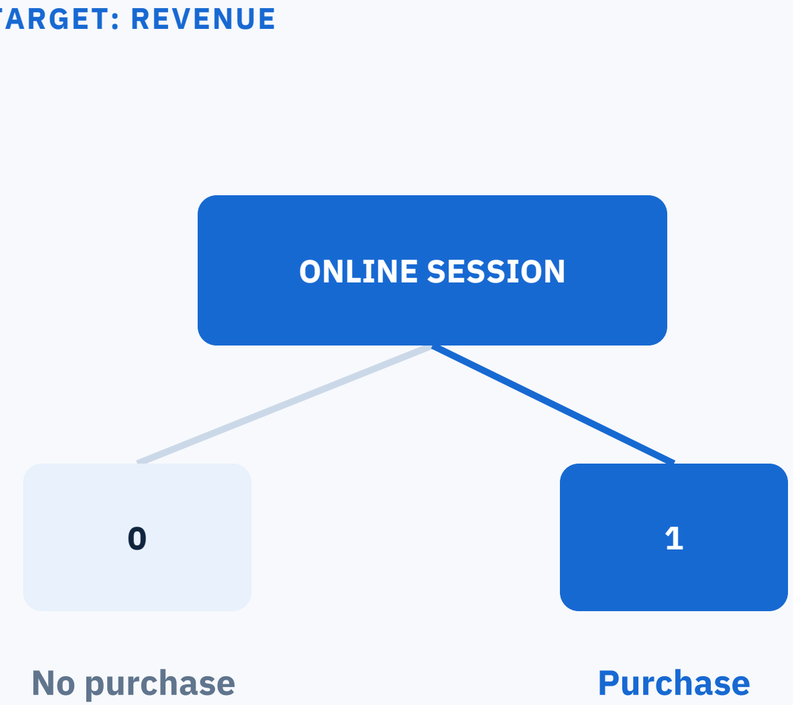


The task

Will this session end in a purchase?

Supervised binary classification of each browsing session, trained offline (batch learning) on labelled data.

Why: help an e-commerce site understand shopping behaviour and act on likely buyers.



The data

12,330 sessions · 17 features + the target Revenue · one row per browsing session

BEHAVIOUR

Pages visited and time spent (Administrative, Informational, ProductRelated) · BounceRates, ExitRates, PageValues

CONTEXT

VisitorType · OperatingSystems · Browser · Region · TrafficType

CALENDAR

Month · Weekend · SpecialDay (closeness to a special day)

TARGET

Revenue (purchase or not): 84.5 % no purchase, 15.5 % purchase → imbalanced classes

Exploration

Buyers behave differently, and they are the minority (15.5 %).

01

Visitor type matters

Purchase rate: 24.9 % for new visitors vs 13.9 % for returning visitors.

02

Strong seasonality

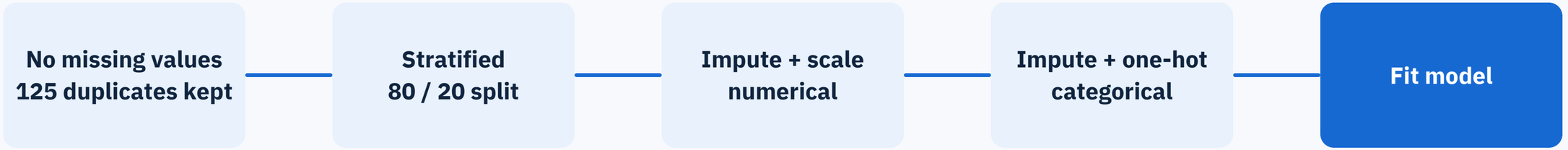
25.4 % of November sessions end in a purchase, only 1.6 % in February.

03

PageValues stands out

Average PageValues 27.3 for buyers vs 2.0; lower bounce (0.005 vs 0.025) and exit rates (0.020 vs 0.047).

Cleaning and preparation



Train: 9,864 sessions · Test: 2,466 sessions · same purchase rate (15.5 %)

Numerical features: median imputation + standard scaling

Categorical (Month, VisitorType, Weekend): most-frequent imputation + one-hot encoding

All steps in one scikit-learn Pipeline (ColumnTransformer), learned on training data only

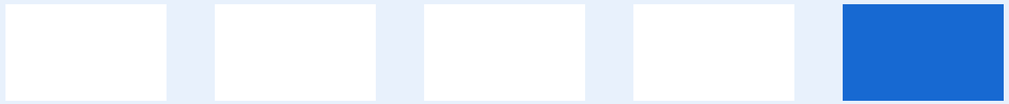
Training protocol

Held-out test set

20 % of the sessions (2,466), stratified.
Used only once, for the final evaluation.

5-fold stratified cross-validation

On the training set only (9,864 sessions), same folds for every model. Baseline: a dummy classifier that always predicts “no purchase”.



Selection metric: average precision (AP), suited to imbalanced classes
Also recorded: accuracy, precision, recall, F1, ROC-AUC

Models and tuning

Logistic regression

Regularization C : 1
Class weights : balanced

Random forest

Number of trees : 200
Depth : 8
Min samples leaf : 15
Max features 0.7
Class weights : None

Gradient boosting

Number of stages : 200
Learning rate : 0.3
Max depth : 3
Min samples leaf : 20
Subsample : 1

Decision tree

Max depth : 8
Min samples leaf : 20
Class weights : None

Validation results

Best classifier: random forest, highest mean validation AP over the 5 folds

MODEL	VALIDATION AP	ROC-AUC	F1
Logistic regression	0.659	0.906	0.630
Random forest (selected)	0.755	0.933	0.665
Gradient boosting	0.751	0.933	0.663
Decision tree	0.690	0.906	0.633

Why: AP is our selection metric for imbalanced data (15.5 % buyers); the random forest also has the best F1. Gradient boosting is a close second.

Test scores

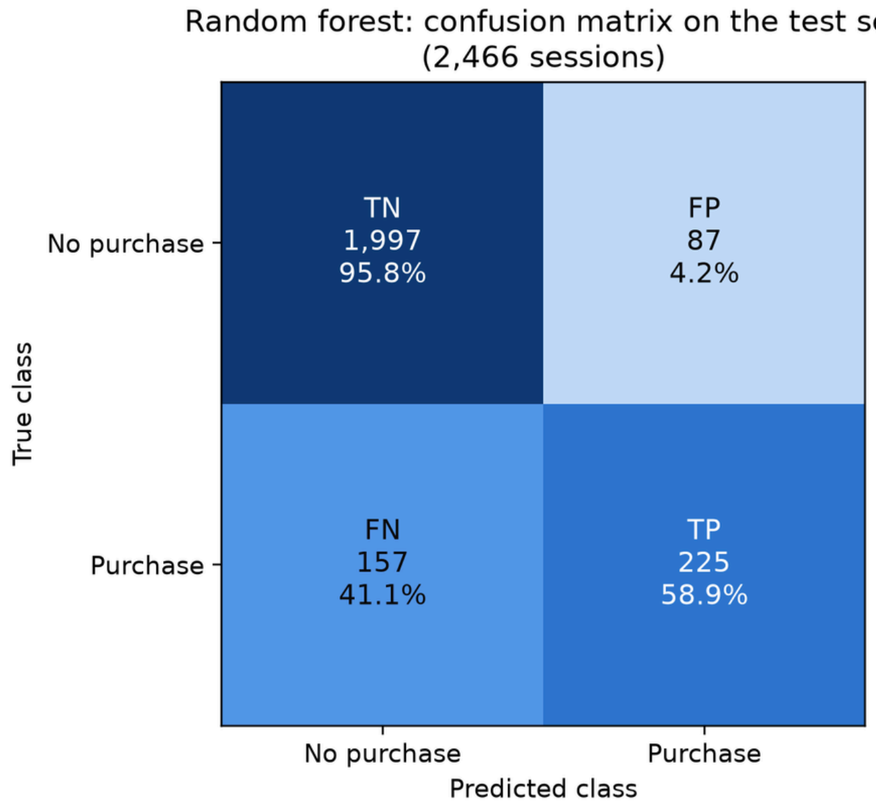
Final evaluation on the 2,466 test sessions

MODEL	PRECISION	RECALL	F1	ROC-AUC	AP
Logistic regression	0.511	0.749	0.607	0.896	0.626
Random forest (selected)	0.721	0.589	0.648	0.930	0.744
Gradient boosting	0.718	0.594	0.650	0.929	0.744
Decision tree	0.712	0.563	0.629	0.908	0.682

Dummy baseline: precision 0 · recall 0 · F1 0 · ROC-AUC 0.500 · AP 0.155 · accuracy 84.5 %
Random forest: validation AP 0.755 → test AP 0.744

Confusion matrix

Selected model · random forest · test set



225 of 382 purchases detected

Recall 58.9 %: 157 purchases missed (FN)

Specificity 95.8 %: only 87 false positive out of 2,084 sessions without purchase

Precision 72.1 %: about 7 in 10 predicted purchases are real

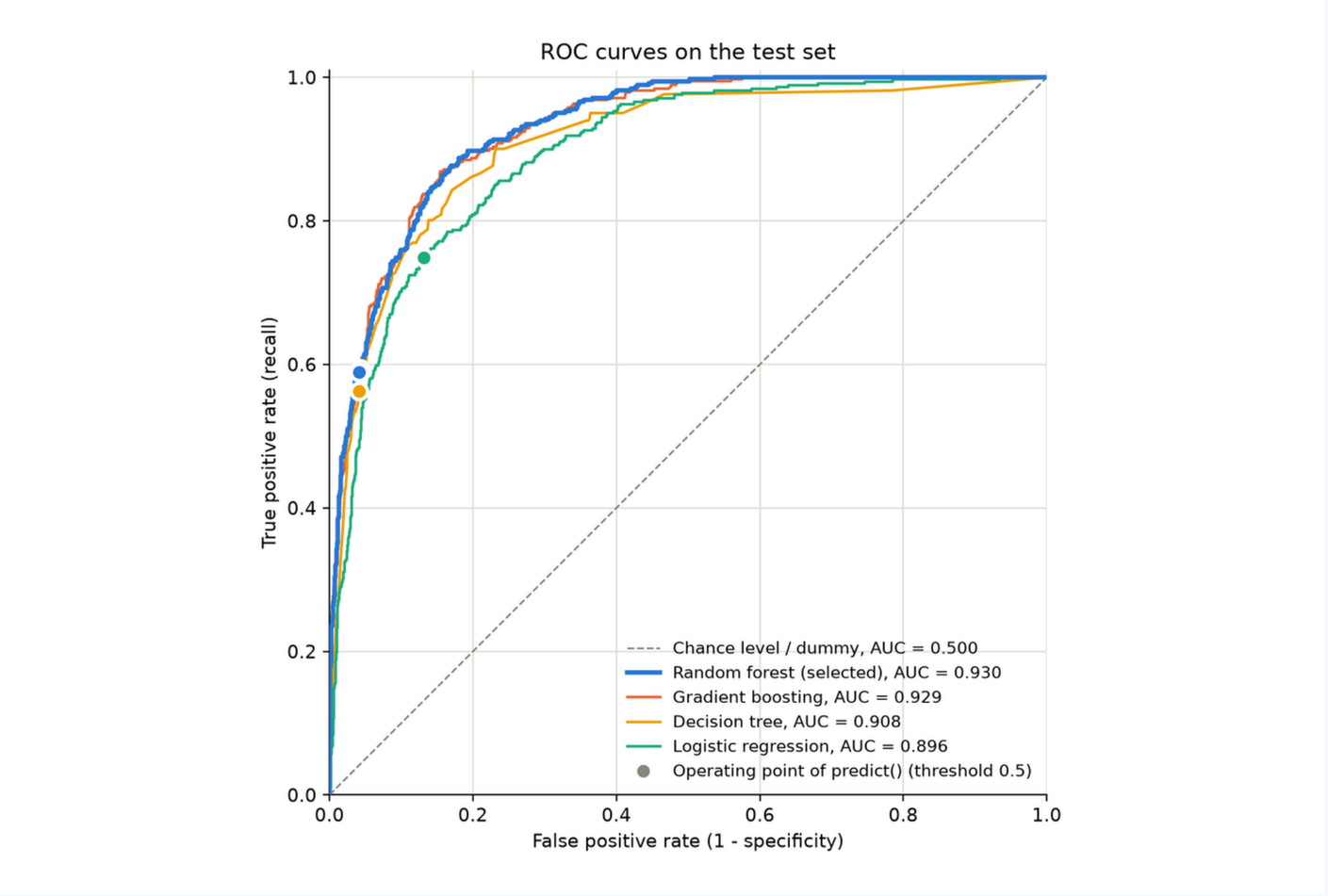
ROC curves

Four tuned models · test set

All models far above chance

AUC: random forest 0.930, gradient boosting 0.929, decision tree 0.908, logistic regression 0.896 (chance: 0.5)

Dots = predict() at threshold 0.5.
The tree-based models sit on the steep part of their curves: a lower threshold would detect more buyers for few extra false positives.



Conclusion

WHAT WE DID

Explored and cleaned 12,330 sessions, built a leakage-safe pipeline, tuned four models by cross-validation and evaluated the best one once on the test set.

WHAT WORKED

Best classifier: random forest, chosen on validation AP and confirmed on the test set (AP 0.744 vs 0.155 for the baseline, ROC-AUC 0.930). Gradient boosting is very close.

LIMITS

Only 59 % of buyers found at threshold 0.5.